

MSSC 6010

**Exam I (Fall 20 25)
Time Limit: 70 minutes**

Section 101

Print Your Name: Solution

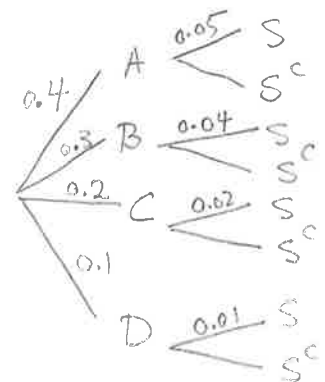
Notes:

- 1 - DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO.**
- 2 - GIVE ALL THE NECESSARY DETAILS TO GET FULL CREDITS.**
- 3 - NO ELECTRONIC DEVICES OTHER THAN A CALCULATOR MAY BE USED.**

20 pts

1. Four different transportation companies (A, B, C, and D) deliver the products of an agricultural firm. Company A transports 40% of the products; company B, 30%; company C, 20%; and, finally, company D, 10%. During transportation, 5%, 4%, 2%, and 1% of the products spoil with companies A, B, C, and D, respectively. If one product is randomly selected,
- Obtain the probability that it is spoiled.
 - If the chosen product is spoiled, derive the probability that company A has transported it.
 - From a truck that belongs to company A and it delivers the products of this agricultural firm, we randomly selected five products, what is the probability that at most one of the products is spoiled?

$$\begin{aligned}
 P(S) &= P(S|A) + \dots + P(S|D) \\
 &= P(S|A)P(A) + \dots + P(S|D)P(D) \\
 &= 0.4(0.05) + 0.3(0.04) + 0.2(0.02) + 0.1(0.01) \\
 &= 0.037 \Rightarrow \boxed{P(S) = 0.037} \text{ (a)}
 \end{aligned}$$



$$\begin{aligned}
 P(A|S) &= \frac{P(S|A)P(A)}{P(S)} = \frac{0.05 \times 0.4}{0.037} \\
 &\Rightarrow \boxed{P(A|S) = 0.54} \text{ (b)}
 \end{aligned}$$

$$X \sim \text{Bin}(n, \pi) \text{ where } n=5 \text{ \& } \pi = P(S|A) = 0.05$$

$$\begin{aligned}
 P(X \leq 1) &= P(X=0) + P(X=1) \\
 &= \frac{5!}{0! 5!} (0.05)^0 (0.95)^5 + \frac{5!}{1! 4!} (0.05)^1 (0.95)^4 \\
 &\Rightarrow \boxed{P(X \leq 1) = 0.9774} \text{ (c)}
 \end{aligned}$$

PLEASE DO NOT WRITE IN THE FOLLOWING SPACE.

1	2	3	4	5	Total

20 pts

2. Let X be a random variable with probability density function

$$f(x) = k e^{-\frac{x}{\theta}}, \quad x \geq 0, \quad \theta > 0.$$

- Find k so that the given function is a valid pdf.
- Find the moment generating function of X .
- Find the cumulative distribution function $F(\cdot)$.
- Let u to be an observation from $Unif(0,1)$, how can we obtain a realization from X using u .

$$\begin{aligned} \textcircled{a} \int_{-\infty}^{\infty} f(x|\theta) dx &= 1 \Rightarrow \int_0^{\infty} k e^{-\frac{x}{\theta}} dx = -k\theta e^{-\frac{x}{\theta}} \Big|_0^{\infty} = 0 - (-k\theta) = 1 \\ &\Rightarrow k = \frac{1}{\theta} \Rightarrow f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad x \geq 0 \end{aligned}$$

$$\begin{aligned} \textcircled{b} M_X(t) &= E(e^{tx}) = \int_0^{\infty} e^{tx} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \frac{1}{\theta} \int_0^{\infty} e^{(t-\frac{1}{\theta})x} dx \\ &= \frac{1}{\theta} \left(t - \frac{1}{\theta} \right)^{-1} e^{(t-\frac{1}{\theta})x} \Big|_0^{\infty} = 0 - \frac{1}{\theta} \left(\frac{t\theta - 1}{\theta} \right)^{-1} \\ &\Rightarrow M_X(t) = (1 - \theta t)^{-1} \quad \text{for } |t| < \frac{1}{\theta} \end{aligned}$$

$$\begin{aligned} \textcircled{c} F(x) &= \int_0^x \frac{1}{\theta} e^{-\frac{t}{\theta}} dt = -e^{-\frac{t}{\theta}} \Big|_0^x = 1 - e^{-\frac{x}{\theta}} \\ &\Rightarrow F(x) = 1 - e^{-\frac{x}{\theta}} \quad \text{for } x \geq 0 \end{aligned}$$

$$\textcircled{d} \text{ Let } u = F(x) \Rightarrow u = 1 - e^{-\frac{x}{\theta}} \Rightarrow e^{-\frac{x}{\theta}} = 1 - u$$

$$\Rightarrow -\frac{x}{\theta} = \ln(1-u) \Rightarrow \boxed{x = -\theta \ln(1-u)}$$

x belongs to $F(\cdot)$

20 pts

3. Suppose that the average arrival rate at a local fast food drive through window is three cars per minute ($\lambda = 3$). Find

- The probability that at most three cars arrive in 90 seconds.
- The probability that less than one minute elapses before the first car arrives.
- The probability that more than two minutes elapses before the third car arrives.
- What is the average waiting time before the third car arrives?

Note: Give the details on the distributions you used in each part to get the full credit.

(a) $X \sim \text{Poisson}(\lambda t)$ where $\lambda = 3$ $t = 1.5$ (90 seconds in min)

$$\Rightarrow X \sim \text{Poisson}(4.5)$$

$$P(X \leq 3) = \sum_{x=0}^3 \frac{(4.5)^x e^{-4.5}}{x!} = \sum_{x=0}^3 \frac{4.5^x e^{-4.5}}{x!} \Rightarrow \boxed{P(X \leq 3) = 0.342}$$

(b) $Y \sim \text{Exp}(\lambda = 3)$ where $f(y) = \lambda e^{-\lambda y}$ $y \geq 0$

$$P(Y < 1) = \int_0^1 \lambda e^{-\lambda y} dy = 1 - e^{-\lambda} = 1 - e^{-3} \Rightarrow \boxed{P(Y < 1) = 0.95}$$

(c) $Z = Y_1 + Y_2 + Y_3 \Rightarrow Z \sim \text{Gamma}(\alpha = 3, \lambda = 3)$ $f(z) = \frac{1}{\Gamma(3)} \lambda^3 z^2 e^{-\lambda z}$

$$P(Z > 2) = 1 - P(Z < 2) = 1 - \int_0^2 \frac{27}{2} z^2 e^{-3z} dz$$

$$\Rightarrow \boxed{P(Z > 2) = 1 - 25e^{-6} = 1 - 0.9380} \Rightarrow \boxed{P(Z > 2) = 0.062}$$

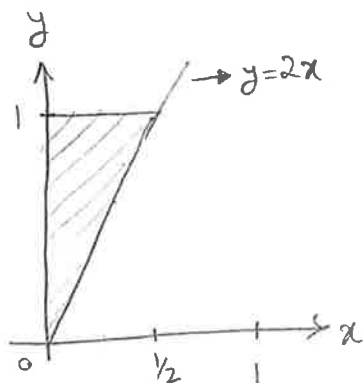
(d) $W \sim \text{Gamma}(\alpha = 3, \lambda = 3)$ $\neq$ memoryless property

$$\boxed{E(W) = \frac{\alpha}{\lambda} = \frac{3}{3} = 1 \text{ a min}}$$

20 pts

4. If $f(x, y) = K(y - 2x)$ is a joint density function over $0 < x < 1$, $0 < y < 1$, and $y > 2x$.

- Find K so that the given function is a valid pdf.
- Find the marginal density of X and the conditional density of $X|Y$.
- Are X and Y independent? Justify.



$$(a) \iint_{-\infty}^{\infty} f(x, y) dx dy = 1 \Rightarrow \int_0^1 \int_0^{y/2} K(y - 2x) dx dy = 1 \Rightarrow$$

$$= K \int_0^1 (yx - x^2) \Big|_{x=0}^{x=y/2} dy$$

$$= K \int_0^1 \left(\frac{y^2}{2} - \frac{y^2}{4} \right) dy = K \int_0^1 \frac{y^2}{4} dy$$

$$= K \frac{y^3}{12} \Big|_{y=0}^{y=1} = K \left(\frac{1}{12} \right)$$

$$\Rightarrow \boxed{K = 12}$$

$$(b) f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^{y/2} 12(y - 2x) dx = 12(yx - x^2) \Big|_{x=0}^{x=y/2} \Rightarrow \boxed{f_Y(y) = 3y^2 \quad 0 < y < 1}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = \int_{2x}^1 12(y - 2x) dy = 12 \left(\frac{y^2}{2} - 2xy \right) \Big|_{y=2x}^{y=1} = 12 \left[\left(\frac{1}{2} - 2x \right) - (2x^2 - 4x^2) \right]$$

$$\Rightarrow \boxed{f_X(x) = 12(2x^2 - 2x + \frac{1}{2}) \quad 0 < x < \frac{1}{2}}$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{12(y - 2x)}{3y^2} \Rightarrow \boxed{f_{X|Y}(y) = 4(y - 2x)y^{-2}, \quad 0 < x < y/2}$$

(c) $f_{X,Y}(x, y) \neq f_X(x)f_Y(y) \Rightarrow X \& Y$ are not independent.

20 pts

5. A particular unfair coin is constructed so that the probability of obtaining a tail is $2/3$. The unfair coin is flipped twice. Define two random variables: Z = the number of heads in the second flip and W = the number of heads in two flips.

- Construct a table showing the joint probability distribution of both random variables Z and W including the marginal probabilities.
- Find the covariance between Z and W . Are they independent?
- Suppose the covariance between Z and W were 0. Would this imply that Z and W are independent?

$$P(T) = 2/3, P(H) = 1/3, \Omega = \{HH, HT, TH, TT\} \Rightarrow \begin{cases} Z = \# \text{ of heads in } 2^{\text{nd}} \text{ flip} = \{0, 1\} \\ W = \# \text{ of heads in 2 flips} = \{0, 1, 2\} \end{cases}$$

(a)

$Z \backslash W$	0	1	2	$P_Z(z)$
0	$(2/3)^2$	$(2/3)(1/3)$	0	$6/9$
1	0	$(1/3)(2/3)$	$(1/3)^2$	$3/9$
$P_W(w)$	$4/9$	$4/9$	$1/9$	Total = 1

$$\Rightarrow P_W(w) = \begin{cases} 4/9 & w=0, 1 \\ 1/9 & w=2 \\ 0 & \text{o/w} \end{cases}$$

$$\& P_Z(z) = \begin{cases} 2/3 & z=0 \\ 1/3 & z=1 \\ 0 & \text{o/w} \end{cases}$$

(b) $\text{Cov}(Z, W) = E(ZW) - E(Z)E(W)$

$$\begin{cases} E(ZW) = 0 \times 0 \times 4/9 + 0 \times 1 \times 2/9 + 0 \times 2 \times 0 + 1 \times 0 \times 0 + 1 \times 1 \times 2/9 + 1 \times 2 \times 1/9 \Rightarrow E(ZW) = 4/9 \\ E(Z) = 0 \times 2/3 + 1 \times 1/3 = 1/3 \Rightarrow E(Z) = 1/3 \\ E(W) = 0 \times 4/9 + 1 \times 4/9 + 2 \times 1/9 = 6/9 \Rightarrow E(W) = 2/3 \end{cases}$$

$$\Rightarrow \text{Cov}(Z, W) = 4/9 - 1/3 \times 2/3 \Rightarrow \text{Cov}(Z, W) = 2/9$$

$$\Rightarrow \text{Cov}(Z, W) \neq 0 \Rightarrow \boxed{Z \& W \text{ are NOT indep.}}$$

(c) No, it is not an "if and only if":

$$(X, Y) \text{ are Indep} \Rightarrow \text{Cov}(X, Y) = 0 \quad \text{But} \quad \text{Cov}(X, Y) = 0 \not\Rightarrow \text{Indep.}$$